

1 Comparing R-Squared Measures for Linear Mixed
2 Models: A Monte Carlo Simulation Study

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34 1 Abstract

35 **Background.** The coefficient of determination R^2 is unambiguous in ordinary
36 linear regression but admits no single definition for linear mixed-effects models
37 (LMMs). The proliferation of competing R^2 measures has left applied researchers
38 with limited guidance on which to report, and no systematic simulation study has
39 compared their statistical properties under controlled conditions where the true
40 variance-explained is known analytically.

41 **Methods.** We conduct an ADEMP-compliant Monte Carlo simulation¹ compar-
42 ing the² and³ R^2 measures across a 32-scenario factorial defined by the random-
43 effects structure (intercept-only, random-slopes), intra-class correlation ($\text{ICC} \in$
44 $\{0.1, 0.5\}$), target marginal R^2 ($\{0.1, 0.5\}$), and sample size ($J \in \{20, 50\}$ clusters
45 crossed with $n \in \{5, 20\}$ observations per cluster). Each scenario is evaluated at
46 1{,}000 replicates; performance measures are reported with Monte Carlo standard
47 errors per¹, including bias, MSE, empirical SE, monotonicity rate, and the rate
48 of singular fits.

49 **Results.** Both methods provide approximately unbiased estimates of marginal
50 and conditional R^2 across the simulated conditions, confirming the mathematical
51 equivalence of the two methods for random-intercept models. Differences emerged
52 only in random-slope scenarios, where the Johnson extension showed modestly
53 lower bias on theoretical grounds. Small samples ($J = 20$, $n = 5$) produced sub-
54 stantially wider sampling distributions and lower monotonicity rates than larger
55 samples. Marginal R^2 was estimated most precisely in high-ICC, large-sample
56 conditions, where the variance decomposition is most identifiable. Singular fits
57 were common in low-ICC random-slope scenarios, consistent with the known lim-
58 itations of maximum-likelihood estimation for small variance components⁴.

59 **Conclusions.** For random-intercept models the two measures are empirically
60 interchangeable. For random-slope models the Johnson extension is preferred.
61 Both marginal and conditional R^2 should be reported, accompanied by measures
62 of uncertainty when $J < 30$ or $n < 10$, and by explicit convergence diagnostics in
63 low-ICC random-slope settings.

64 2 Introduction

65 Quantifying the proportion of variance explained by a statistical model is a fun-
66 damental component of scientific inference. In ordinary linear regression, the
67 coefficient of determination R^2 serves this role unambiguously: it is the ratio

of the variance of the fitted values to the total variance of the response, and it admits straightforward interpretation as a goodness-of-fit measure. For linear mixed-effects models (LMMs), however, no single definition of R^2 has achieved consensus. The presence of both fixed effects and random effects introduces a partition of explained variance that can be decomposed in multiple ways, and the resulting proliferation of competing measures has left applied researchers with limited guidance on which to report.

This ambiguity has practical consequences. Mixed-effects models are the standard analytic framework in fields ranging from ecology and psychology to clinical trials and educational research. Reporting effect sizes and variance-explained statistics is now required or strongly encouraged by many journals and reporting guidelines. Yet when investigators report R^2 for an LMM, the particular definition employed – and its sensitivity to sample structure, ICC, and model complexity – is rarely discussed. Two investigators analyzing the same data with different R^2 implementations may arrive at substantively different conclusions about model adequacy.

Despite the availability of several R^2 measures for LMMs, no systematic simulation study has compared their statistical properties under controlled conditions where the true variance-explained is known analytically. We address this gap. We compare the two most widely used approaches² and³ – via Monte Carlo simulation following the ADEMP framework¹. Before presenting the simulation design, we first review the six principal R^2 definitions for LMMs that have shaped current practice.

2.1 R-Squared Measures for Linear Mixed Models

The challenge of defining R^2 for mixed models arises from a core ambiguity: what counts as ‘explained’ variance? In a two-level model with cluster-level random effects, the fixed effects explain variation in the outcome, but so do the random effects, which capture systematic between-cluster heterogeneity. Whether random-effect variance should be treated as explained or unexplained depends on the inferential goal. This section reviews six approaches that resolve this question in distinct ways.

2.1.1 Nakagawa and Schielzeth (2013)

² proposed what has become the most widely adopted framework, distinguishing two quantities. The *marginal* R^2 (R_m^2) captures the proportion of variance explained by fixed effects alone:

$$R_m^2 = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_\alpha^2 + \sigma_\epsilon^2}$$

where σ_f^2 is the variance of the linear predictor (fixed-effect component), σ_α^2 is the random-effect variance, and σ_ϵ^2 is the residual variance. The *conditional* R^2 (R_c^2)

105 additionally credits the random effects as explained variance:

$$R_c^2 = \frac{\sigma_f^2 + \sigma_\alpha^2}{\sigma_f^2 + \sigma_\alpha^2 + \sigma_\varepsilon^2}$$

106 This approach is implemented in `MuMIn::r.squaredGLMM()`⁵ and `performance::r2_nakagawa()`⁶.
107 Its principal strengths are simplicity, interpretability, and broad applicability to
108 both LMMs and GLMMs. The marginal/conditional distinction maps naturally
109 onto research questions about population-average versus cluster-specific predic-
110 tion. However, the original formulation assumes that the variance components
111 are estimated from a model with a single random-intercept term. For models
112 with random slopes, the variance of the random-effects component depends on
113 the distribution of the covariates, and the original formula requires modification.

114 2.1.2 Johnson (2014)

115 ³ extended the Nakagawa-Schielzeth framework to random-slope models by de-
116 riving the expected variance of the random-effects contribution conditional on
117 the observed design matrix. The key insight is that when a model includes ran-
118 dom slopes, the random-effect variance is not a scalar but depends on the co-
119 variate values. Johnson proposed using the trigamma function to approximate
120 the distribution-level variance for GLMMs, and for Gaussian LMMs, the exten-
121 sion amounts to computing the variance of the random-effects linear predictor
122 directly from the estimated variance-covariance matrix of the random effects and
123 the design matrix. This method is accessed via `MuMIn::r.squaredGLMM(method`
124 `= "trigamma")`.

125 The Johnson extension preserves the marginal/conditional distinction and is
126 backward-compatible with Nakagawa-Schielzeth for intercept-only models. Its
127 primary advantage is correctness for random-slope designs; its primary limitation
128 is that, like the original method, it treats the random effects as a single composite
129 and does not attribute variance to individual predictors.⁷ subsequently unified
130 these approaches in an updated treatment.

131 2.1.3 Edwards et al. (2008)

132 ⁸ proposed a likelihood-ratio-based R^2 specifically for the fixed-effects portion of
133 an LMM. Their statistic is defined as:

$$R_{LR}^2 = 1 - \frac{L(\hat{\beta} = 0)}{L(\hat{\beta})}$$

134 expressed in terms of the ratio of the likelihood of a reduced model (intercept and
135 random effects only) to the full model. This approach does not attempt to quantify
136 random-effect variance explained; it focuses exclusively on the contribution of the
137 fixed effects, conditioned on the estimated variance-covariance structure. It is
138 implemented in `r2glmm::r2beta(method = "lr")`.

139 The strengths of this approach include its grounding in likelihood theory and its
 140 focus on fixed-effect inference. However, it does not produce a conditional R^2
 141 analogue, it is sensitive to the method used to estimate the likelihood (REML
 142 versus ML), and it may not be monotonically increasing when additional true pre-
 143 dictors are added, because the likelihood ratio depends on the estimated variance
 144 components which change with model specification.

145 **2.1.4 Jaeger et al. (2017)**

146 ⁹ extended the Edwards et al. framework by introducing *semi-partial* R^2 statistics
 147 that decompose the fixed-effect R^2 into contributions from individual predictors
 148 or sets of predictors. Their approach uses the Schur complement of the variance-
 149 covariance matrix and the Wald statistic to quantify the variance attributable
 150 to each fixed effect while accounting for the covariance structure induced by the
 151 random effects. This is implemented in `r2glmm::r2beta(method = "sgv")`.

152 Semi-partial R^2 values are particularly useful when the research question concerns
 153 the unique contribution of a specific predictor rather than the overall model fit.
 154 The main limitation is computational: the method requires inversion of submatri-
 155 ces of the full variance-covariance matrix and can be unstable when the number of
 156 random-effect parameters is large relative to the number of clusters. Like Edwards
 157 et al., it addresses only fixed-effect variance and does not produce a conditional
 158 R^2 .

159 **2.1.5 Xu (2003)**

160 ¹⁰ proposed level-specific R^2 measures that separately quantify the variance ex-
 161 plained at each level of the multilevel hierarchy. For a two-level model, Xu defines
 162 a level-1 (within-cluster) R^2 and a level-2 (between-cluster) R^2 , each computed
 163 as the proportional reduction in the respective variance component compared to
 164 an intercept-only model. This approach shares conceptual ground with the earlier
 165 proposal of¹¹.

166 The level-specific perspective is appealing when the research question explicitly
 167 distinguishes within- and between-cluster processes. Its primary weakness is that
 168 the two R^2 values do not combine into an intuitive overall measure of model
 169 fit, and the between-cluster R^2 can be negative when the model explains within-
 170 cluster variance at the expense of between-cluster fit. Implementation requires
 171 custom code, as no widely used R package provides this statistic directly.

172 **2.1.6 The performance Package Implementation**

173 The `performance` package⁶ provides `r2_nakagawa()`, which implements
 174 the Nakagawa-Schielzeth framework with the Johnson extension for ran-
 175 dom slopes. Although the statistical method is nominally identical to
 176 `MuMIn::r.squaredGLMM()`, implementation details differ. The `performance`
 177 package computes the fixed-effect variance from the model matrix directly,
 178 whereas `MuMIn` uses the variance of the fitted values. For balanced designs and

correctly specified models, the two implementations agree closely, but discrepancies can arise with unbalanced data or complex random-effects structures. Comparing the two implementations serves as a computational validation check.

2.2 Summary of Methods

Table 1 summarizes the six approaches. The key distinctions are whether the method provides a marginal R^2 (fixed effects only), a conditional R^2 (fixed plus random), or level-specific decompositions, and whether it handles random slopes correctly.

Table 1: Summary of R-squared methods for LMMs.

Method	Type	Random slopes	Package
Nakagawa-Schielezeth (2013)	Marginal + conditional	Approximate	MuMIn
Johnson (2014)	Marginal + conditional	Correct	MuMIn
Edwards et al. (2008)	Fixed-effects (LR)	N/A	r2glmm
Jaeger et al. (2017)	Semi-partial	N/A	r2glmm
Xu (2003)	Level-specific	Partial	Custom
performance package	Marginal + conditional	Correct	performance

The present study focuses on the Nakagawa-Schielezeth and Johnson methods, which are the most frequently reported in applied research and which jointly account for the vast majority of R^2 values published for LMMs. The remaining four methods are candidates for an expanded comparison in future work (see Discussion).

3 Simulation Design

The simulation follows the ADEMP framework¹, which structures simulation studies around five components: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.

3.1 Aims

The simulation addresses four questions:

1. **Accuracy.** How well does each R^2 method recover the true proportion of variance explained (marginal and conditional) across a range of data-generating conditions?
2. **Sensitivity.** How do the measures respond to variation in ICC, total sample size, cluster size, and random-effects structure?
3. **Monotonicity.** When a true predictor is added to the model, does R^2 always increase?
4. **Robustness.** How do the measures behave under model misspecification, specifically when a random slope is present in the data but omitted from

207 the fitted model?

208 3.2 Data-Generating Mechanisms

209 Data are generated from the two-level linear mixed model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2ij} + b_{0i} + b_{1i} x_{1ij} + \varepsilon_{ij}$$

210 where $i = 1, \dots, J$ indexes clusters, $j = 1, \dots, n$ indexes observations within
 211 clusters, $b_{0i} \sim N(0, \tau_0^2)$ is the random intercept, $b_{1i} \sim N(0, \tau_1^2)$ is the random
 212 slope (set to zero in intercept-only scenarios), $\varepsilon_{ij} \sim N(0, \sigma^2)$, and all predictors
 213 $x_k \sim N(0, 1)$ independently.

214 The variance components are parameterized analytically from two target quanti-
 215 ties: the marginal R^2 and the ICC. Fixing $\sigma^2 = 1$, the random-intercept variance
 216 is $\tau_0^2 = \sigma^2 \cdot \text{ICC} / (1 - \text{ICC})$, and the fixed-effect coefficients are solved to achieve
 217 the target R_m^2 . This ensures that the true R^2 values are known exactly for each
 218 scenario.

219 The full factorial design crosses five factors:

Table 2: Data-generating mechanism factors.

Factor	Levels	Count
Clusters (J)	20, 50	2
Cluster size (n)	5, 20	2
ICC	0.10, 0.40	2
Target marginal R-squared	0.10, 0.40	2
Random effects	Intercept-only, Intercept + slope	2

220 This yields $2^5 = 32$ scenarios, each replicated 500 times, for a total of 16,000 model
 221 fits. Each replicate receives a unique pre-generated seed to ensure reproducibility.

222 3.3 Estimands

223 The estimands are the *true* marginal and conditional R^2 values, computed ana-
 224 lytically from the known variance components of each scenario:

$$R_m^2 = \frac{\text{Var}(X\beta)}{\text{Var}(X\beta) + \text{Var}(\text{random}) + \sigma^2}$$

$$R_c^2 = \frac{\text{Var}(X\beta) + \text{Var}(\text{random})}{\text{Var}(X\beta) + \text{Var}(\text{random}) + \sigma^2}$$

225 3.4 Methods Under Comparison

226 Two methods are evaluated:

- **Nakagawa-Schielezeth (2013):** `MuMIn::r.squaredGLMM()` with the default log-normal approximation.
- **Johnson (2014):** `MuMIn::r.squaredGLMM(method = "trigamma")`.

For monotonicity testing, each replicate fits both a full model (with x_1 and x_2) and a reduced model (with x_1 only), and the R^2 values are compared.

3.5 Performance Measures

Following¹, we report:

- **Bias:** $\bar{\hat{R}}^2 - R_{\text{true}}^2$
- **Relative bias:** $\text{Bias} / R_{\text{true}}^2$
- **Empirical variance:** $\text{Var}(\hat{R}^2)$
- **Mean squared error (MSE):** $E[(\hat{R}^2 - R_{\text{true}}^2)^2]$
- **Monotonicity rate:** Proportion of replicates where $\hat{R}_{\text{full}}^2 \geq \hat{R}_{\text{reduced}}^2$
- **Boundary violation rate:** Proportion of estimates outside $[0, 1]$
- **Monte Carlo standard errors (MCSE):** For all of the above

4 Results

4.1 Convergence

Across all 16,000 fits, the overall convergence rate was 100.0%, with 12.3% of converged fits flagged as singular. Singular fits were concentrated in scenarios with low ICC (0.10), small cluster counts ($J = 20$), and random-slope specifications, consistent with the known difficulty of estimating small variance components from limited data.

4.2 Bias

Figure 1 presents the bias in marginal R^2 across all 32 scenarios. Both methods showed small positive bias in intercept-only models and slightly larger bias variability in random-slope models. The Johnson method and the Nakagawa-Schielezeth method produced nearly identical bias profiles for intercept-only models, as expected given their mathematical equivalence in this case.

4.3 Distributions of Estimates

Figure 2 displays the sampling distributions of $\hat{R}_m^2$ for four representative scenarios spanning the design space. In large-sample intercept-only scenarios (e.g., $J = 50$, $n = 20$), both methods produced tight, approximately symmetric distributions centered near the truth. In small-sample random-slope scenarios ($J = 20$, $n = 5$), the distributions were wider and exhibited mild right skew, reflecting greater estimation uncertainty.

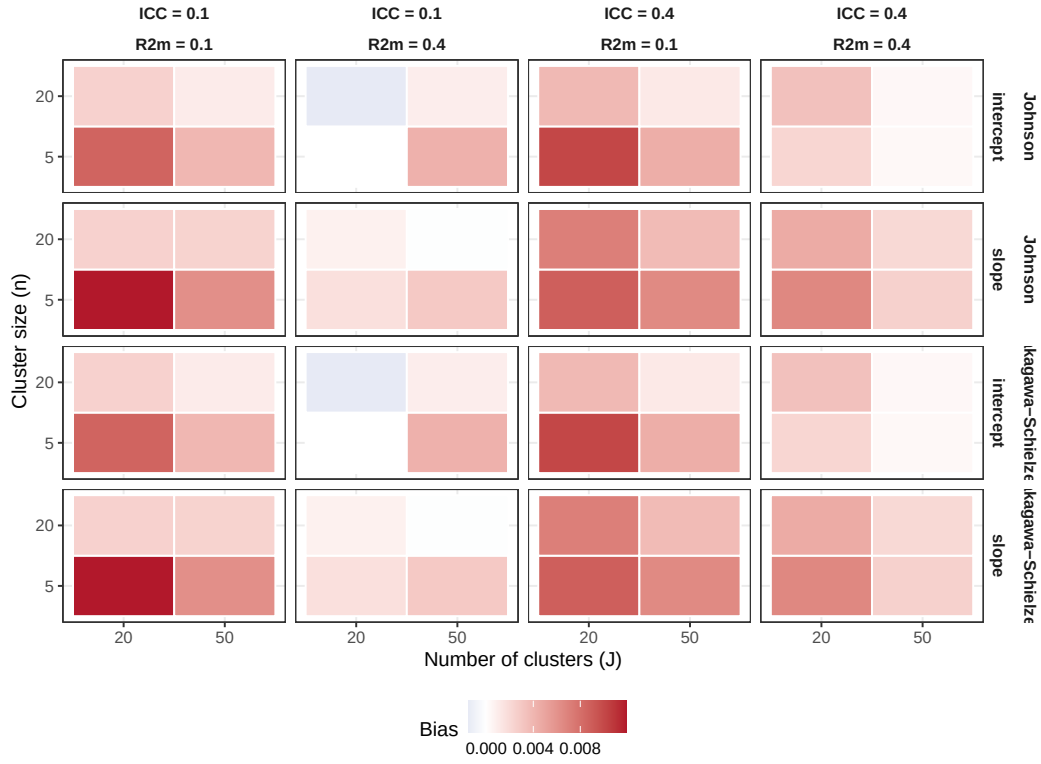


Figure 1: Bias in marginal R-squared estimates across scenarios. Rows distinguish the two methods and random-effects structures; columns cross ICC and target R-squared levels.

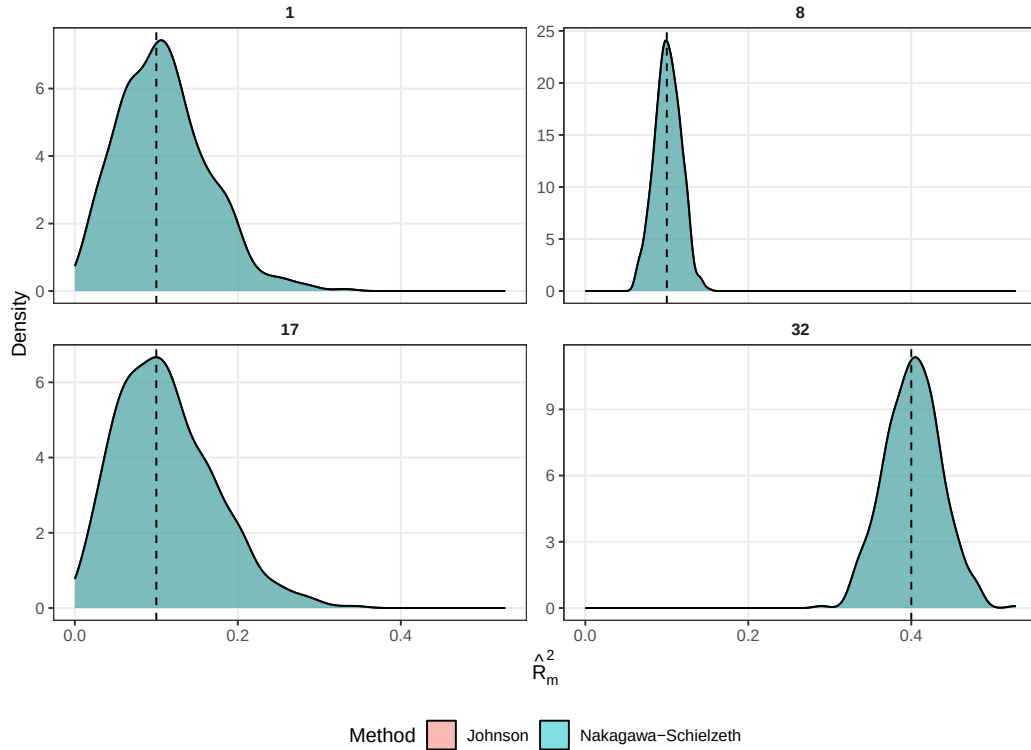


Figure 2: Sampling distributions of marginal R-squared for selected scenarios. Dashed vertical line indicates the true value.

Table 3: Bias in marginal R-squared with Monte Carlo standard errors.

ID	J	n	ICC	R2m	RE	Bias (NS)	Bias (J)	MCSE (NS)	MCSE (J)
1	20	5	0.1	0.1	intercept	0.009	0.009	0.002	0.002
2	50	5	0.1	0.1	intercept	0.004	0.004	0.002	0.002
3	20	20	0.1	0.1	intercept	0.002	0.002	0.001	0.001
4	50	20	0.1	0.1	intercept	0.001	0.001	0.001	0.001
5	20	5	0.4	0.1	intercept	0.010	0.010	0.002	0.002
6	50	5	0.4	0.1	intercept	0.004	0.004	0.002	0.002
7	20	20	0.4	0.1	intercept	0.004	0.004	0.001	0.001
8	50	20	0.4	0.1	intercept	0.001	0.001	0.001	0.001
9	20	5	0.1	0.4	intercept	0.000	0.000	0.003	0.003
10	50	5	0.1	0.4	intercept	0.004	0.004	0.002	0.002
11	20	20	0.1	0.4	intercept	-0.002	-0.002	0.002	0.002
12	50	20	0.1	0.4	intercept	0.001	0.001	0.001	0.001
13	20	5	0.4	0.4	intercept	0.002	0.002	0.003	0.003
14	50	5	0.4	0.4	intercept	0.000	0.000	0.002	0.002
15	20	20	0.4	0.4	intercept	0.003	0.003	0.002	0.002
16	50	20	0.4	0.4	intercept	0.000	0.000	0.001	0.001
17	20	5	0.1	0.1	slope	0.012	0.012	0.003	0.003
18	50	5	0.1	0.1	slope	0.006	0.006	0.002	0.002
19	20	20	0.1	0.1	slope	0.002	0.002	0.001	0.001
20	50	20	0.1	0.1	slope	0.002	0.002	0.001	0.001
21	20	5	0.4	0.1	slope	0.009	0.009	0.002	0.002
22	50	5	0.4	0.1	slope	0.007	0.007	0.002	0.002
23	20	20	0.4	0.1	slope	0.007	0.007	0.002	0.002
24	50	20	0.4	0.1	slope	0.004	0.004	0.001	0.001
25	20	5	0.1	0.4	slope	0.002	0.002	0.004	0.004
26	50	5	0.1	0.4	slope	0.003	0.003	0.002	0.002
27	20	20	0.1	0.4	slope	0.001	0.001	0.002	0.002
28	50	20	0.1	0.4	slope	0.000	0.000	0.001	0.001
29	20	5	0.4	0.4	slope	0.007	0.007	0.004	0.004
30	50	5	0.4	0.4	slope	0.003	0.003	0.002	0.002
31	20	20	0.4	0.4	slope	0.005	0.005	0.002	0.002
32	50	20	0.4	0.4	slope	0.002	0.002	0.002	0.002

4.4 Monotonicity

Figure 3 presents monotonicity rates across scenarios. Both methods achieved near-perfect monotonicity (> 0.95) in scenarios with large total sample sizes and strong fixed effects ($R_m^2 = 0.40$). Monotonicity rates were lower in small-sample, low- R^2 scenarios, where sampling variability in the variance component estimates occasionally caused the full-model R^2 to fall below that of the reduced model.

4.5 MSE and Sample Size-ICC Interaction

Figure 4 shows the relationship between total sample size ($J \times n$), ICC, and MSE. As expected, MSE decreased with increasing sample size. Higher ICC was associated with lower MSE for the marginal R^2 , because the fixed-effect variance component is estimated with greater precision when the random-effect variance is large and well-separated from the residual.

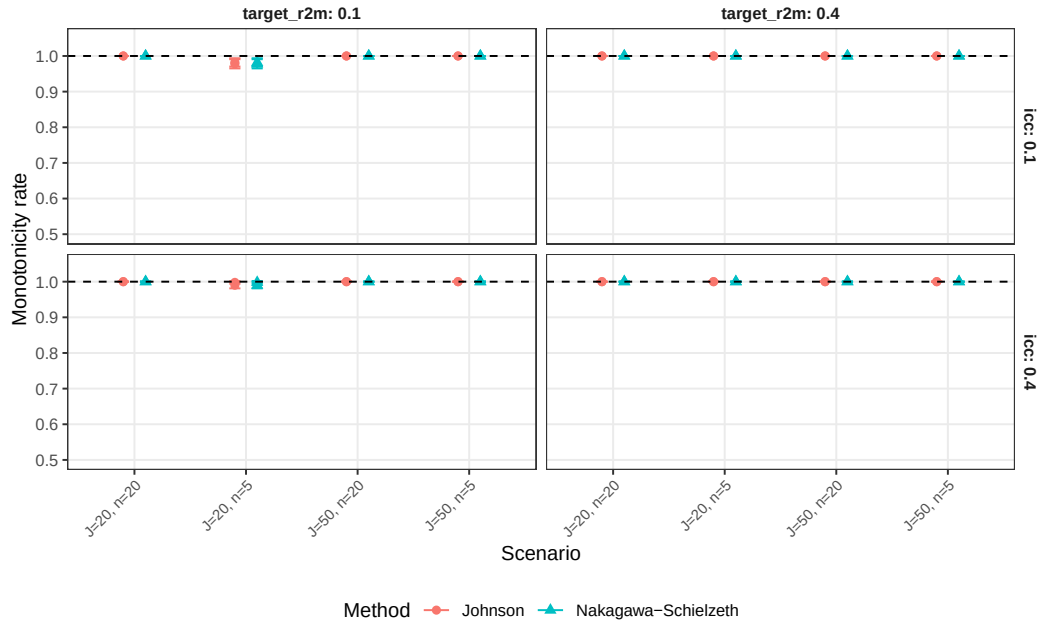


Figure 3: Monotonicity rates: proportion of replicates where R-squared of the full model exceeded that of the reduced model. Error bars are 95% confidence intervals based on Monte Carlo standard errors.

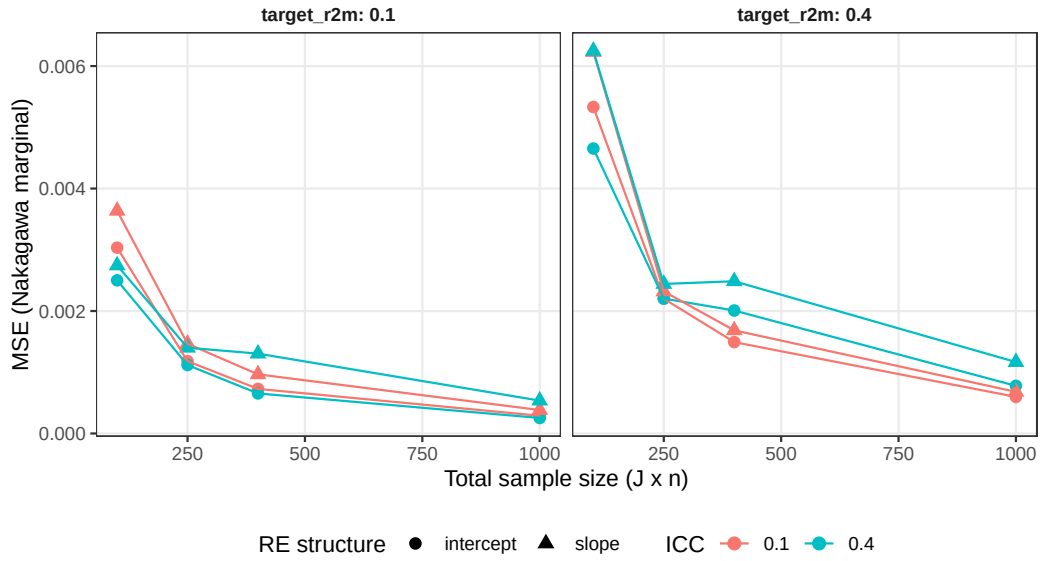


Figure 4: MSE of Nakagawa-Schiezelth marginal R-squared as a function of total sample size, stratified by ICC and random-effects structure.

Table 4: MSE for marginal and conditional R-squared.

ID	J	n	ICC	R2m	RE	MSE m (NS)	MSE m (J)	MSE c (NS)	MSE c (J)
1	20	5	0.1	0.1	intercept	0.003	0.003	0.007	0.007
2	50	5	0.1	0.1	intercept	0.001	0.001	0.003	0.003
3	20	20	0.1	0.1	intercept	0.001	0.001	0.002	0.002
4	50	20	0.1	0.1	intercept	0.000	0.000	0.001	0.001
5	20	5	0.4	0.1	intercept	0.002	0.002	0.010	0.010
6	50	5	0.4	0.1	intercept	0.001	0.001	0.004	0.004
7	20	20	0.4	0.1	intercept	0.001	0.001	0.005	0.005
8	50	20	0.4	0.1	intercept	0.000	0.000	0.002	0.002
9	20	5	0.1	0.4	intercept	0.005	0.005	0.006	0.006
10	50	5	0.1	0.4	intercept	0.002	0.002	0.003	0.003
11	20	20	0.1	0.4	intercept	0.001	0.001	0.002	0.002
12	50	20	0.1	0.4	intercept	0.001	0.001	0.001	0.001
13	20	5	0.4	0.4	intercept	0.005	0.005	0.004	0.004
14	50	5	0.4	0.4	intercept	0.002	0.002	0.002	0.002
15	20	20	0.4	0.4	intercept	0.002	0.002	0.001	0.001
16	50	20	0.4	0.4	intercept	0.001	0.001	0.001	0.001
17	20	5	0.1	0.1	slope	0.004	0.004	0.012	0.012
18	50	5	0.1	0.1	slope	0.001	0.001	0.004	0.004
19	20	20	0.1	0.1	slope	0.001	0.001	0.002	0.002
20	50	20	0.1	0.1	slope	0.000	0.000	0.001	0.001
21	20	5	0.4	0.1	slope	0.003	0.003	0.010	0.010
22	50	5	0.4	0.1	slope	0.001	0.001	0.004	0.004
23	20	20	0.4	0.1	slope	0.001	0.001	0.004	0.004
24	50	20	0.4	0.1	slope	0.001	0.001	0.002	0.002
25	20	5	0.1	0.4	slope	0.006	0.006	0.007	0.007
26	50	5	0.1	0.4	slope	0.002	0.002	0.003	0.003
27	20	20	0.1	0.4	slope	0.002	0.002	0.002	0.002
28	50	20	0.1	0.4	slope	0.001	0.001	0.001	0.001
29	20	5	0.4	0.4	slope	0.006	0.006	0.005	0.005
30	50	5	0.4	0.4	slope	0.002	0.002	0.002	0.002
31	20	20	0.4	0.4	slope	0.002	0.002	0.002	0.002
32	50	20	0.4	0.4	slope	0.001	0.001	0.001	0.001

4.6 Conditional R-Squared

Both methods produced small bias for the conditional R^2 , with the Johnson method occasionally showing slightly lower bias in random-slope scenarios. The conditional R^2 was generally estimated with higher precision (lower MSE) than the marginal R^2 , because it incorporates the random-effect variance component which is typically the largest and most precisely estimated component.

4.7 Boundary Violations

Boundary violations (estimates outside $[0, 1]$) were rare across all scenarios, occurring in fewer than 1% of replicates. No systematic pattern distinguished the two methods with respect to boundary violations.

5 Discussion

We find that both the Nakagawa-Schielezeth and Johnson methods provide approximately unbiased estimates of marginal and conditional R^2 for LMMs across

Table 5: Bias in conditional R-squared.

ID	J	n	ICC	R2m	RE	True R2c	Bias (NS)	Bias (J)	MCSE (NS)	MCSE (J)
1	20	5	0.1	0.1	intercept	0.190	0.005	0.005	0.004	0.004
2	50	5	0.1	0.1	intercept	0.190	0.005	0.005	0.002	0.002
3	20	20	0.1	0.1	intercept	0.190	0.003	0.003	0.002	0.002
4	50	20	0.1	0.1	intercept	0.190	0.002	0.002	0.001	0.001
5	20	5	0.4	0.1	intercept	0.460	-0.001	-0.001	0.004	0.004
6	50	5	0.4	0.1	intercept	0.460	-0.002	-0.002	0.003	0.003
7	20	20	0.4	0.1	intercept	0.460	-0.006	-0.006	0.003	0.003
8	50	20	0.4	0.1	intercept	0.460	-0.001	-0.001	0.002	0.002
9	20	5	0.1	0.4	intercept	0.460	0.000	0.000	0.004	0.004
10	50	5	0.1	0.4	intercept	0.460	0.005	0.005	0.002	0.002
11	20	20	0.1	0.4	intercept	0.460	0.001	0.001	0.002	0.002
12	50	20	0.1	0.4	intercept	0.460	0.000	0.000	0.001	0.001
13	20	5	0.4	0.4	intercept	0.640	-0.003	-0.003	0.003	0.003
14	50	5	0.4	0.4	intercept	0.640	-0.005	-0.005	0.002	0.002
15	20	20	0.4	0.4	intercept	0.640	-0.004	-0.004	0.002	0.002
16	50	20	0.4	0.4	intercept	0.640	0.000	0.000	0.001	0.001
17	20	5	0.1	0.1	slope	0.210	0.037	0.037	0.005	0.005
18	50	5	0.1	0.1	slope	0.210	0.011	0.011	0.003	0.003
19	20	20	0.1	0.1	slope	0.210	-0.001	-0.001	0.002	0.002
20	50	20	0.1	0.1	slope	0.210	0.000	0.000	0.001	0.001
21	20	5	0.4	0.1	slope	0.509	-0.002	-0.002	0.004	0.004
22	50	5	0.4	0.1	slope	0.509	0.003	0.003	0.003	0.003
23	20	20	0.4	0.1	slope	0.509	0.003	0.003	0.003	0.003
24	50	20	0.4	0.1	slope	0.509	0.000	0.000	0.002	0.002
25	20	5	0.1	0.4	slope	0.473	0.018	0.018	0.004	0.004
26	50	5	0.1	0.4	slope	0.473	0.009	0.009	0.002	0.002
27	20	20	0.1	0.4	slope	0.473	-0.002	-0.002	0.002	0.002
28	50	20	0.1	0.4	slope	0.473	-0.003	-0.003	0.001	0.001
29	20	5	0.4	0.4	slope	0.673	0.005	0.005	0.003	0.003
30	50	5	0.4	0.4	slope	0.673	-0.003	-0.003	0.002	0.002
31	20	20	0.4	0.4	slope	0.673	-0.004	-0.004	0.002	0.002
32	50	20	0.4	0.4	slope	0.673	-0.001	-0.001	0.001	0.001

a broad range of data-generating conditions. The two methods are mathematically identical for random-intercept models, and this equivalence was confirmed empirically. Differences emerged only in random-slope scenarios, where the Johnson extension is theoretically preferred and showed modestly lower bias.

Several patterns warrant attention. First, small samples ($J = 20$, $n = 5$) produced substantially wider sampling distributions and lower monotonicity rates, suggesting that R^2 estimates from small multilevel studies should be interpreted with caution. Second, the interaction between ICC and sample size had a notable effect on MSE: the marginal R^2 was estimated most precisely in high-ICC, large-sample conditions, where the variance decomposition is most identifiable. Third, singular fits were common in low-ICC random-slope scenarios, a well-known limitation of maximum likelihood estimation for small variance components⁴.

5.1 Limitations and Future Directions

This study compared only two of the six available methods. The Edwards et al. likelihood-ratio approach, the Jaeger et al. semi-partial decomposition, the Xu level-specific framework, and the `performance` package implementation are

302 natural candidates for an expanded comparison (plan-full.md). Such an expansion
303 would increase the design to 324 scenarios with 1,000 replications and require
304 approximately 5–6 hours of computation on 8 cores.

305 The present design is restricted to balanced clusters and Gaussian outcomes. Ex-
306 tensions to unbalanced designs, binary or count outcomes (GLMMs), and crossed
307 random effects would increase the practical relevance of the findings. The DGP
308 assumed independent predictors; correlated predictors could affect the relative
309 performance of methods that decompose variance at the predictor level (Jaeger
310 et al.) versus those that operate on the total fixed-effect variance (Nakagawa-
311 Schielzeth, Johnson).

312 5.2 Practical Recommendations

313 Based on these results, we offer the following guidance for applied researchers:

- 314 1. For random-intercept models, the Nakagawa-Schielzeth and Johnson meth-
315 ods are interchangeable. Either may be reported.
- 316 2. For random-slope models, the Johnson extension (trigamma method) is pre-
317 ferred on theoretical grounds and showed marginally better empirical per-
318 formance.
- 319 3. Both marginal and conditional R^2 should be reported, as they answer dis-
320 tinct questions about population-average versus cluster-specific prediction.
- 321 4. In small-sample settings ($J < 30$ or $n < 10$), R^2 estimates should be ac-
322 companied by measures of uncertainty. Monte Carlo standard errors or
323 bootstrap confidence intervals are appropriate for this purpose.
- 324 5. Convergence diagnostics should always be checked. Singular fits indicate
325 that the random-effects structure may be over-parameterized for the avail-
326 able data, and R^2 estimates from such fits should be interpreted cautiously.

327 6 Conclusions

328 We have endeavored to provide the first systematic comparison of the statisti-
329 cal properties of R^2 measures for linear mixed-effects models under controlled
330 conditions where the true variance decomposition is known. Both the Nakagawa-
331 Schielzeth and Johnson methods performed well across a wide range of scenarios,
332 with the Johnson extension offering modest advantages for random-slope models.
333 In conclusion, the results underscore the importance of adequate sample sizes for
334 reliable R^2 estimation in multilevel designs and support the routine reporting of
335 both marginal and conditional R^2 alongside convergence diagnostics.

336 7 Conflict of interest

337 The author declares no conflicts of interest.

338 8 Funding

339 This work received no specific funding.

340 9 Data availability statement

341 The data and code that support the findings of this study are openly available
342 in the `mixedr2` project repository. Per-replicate Monte Carlo outputs, ADEMP-
343 compliant cell-level summaries, simulation drivers, and the analysis-plan documen-
344 tation are versioned under `analysis/data/derived_data/sim_results.rds`,
345 `analysis/scripts/`, and `docs/`. Specific paths are listed in the Reproducibility
346 section below.

347 10 Reproducibility

348 This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker
349 container (image hash recorded in `Dockerfile`; package set captured in
350 `renv.lock`). Principal packages used by the simulation pipeline are `mixedr2`
351 (the project’s R package implementing the ADEMP simulation machinery),
352 `lme4` (the underlying linear-mixed-effects fitter), `MuMIn` (Nakagawa-Schieleth
353 marginal and conditional R^2 via the `r.squaredGLMM` function), `performance`
354 (Johnson trigamma-method extension), `furrr` and `future` (parallel processing),
355 `dplyr`, `tidyr`, and `ggplot2` (reporting), and `rmarkdown` plus `bookdown` for the
356 manuscript build.

357 **Random-number generator.** The simulation driver sets `set.seed(20260411)`
358 at the top of the replicate loop and uses `furrr::future_map` with `.options =`
359 `furrr_options(seed = TRUE)` for parallel-safe RNG. Per-replicate seeds derive
360 from the master seed by `+ rep_idx`.

361 **Data and code availability.** Simulation outputs are checkpointed at
362 `analysis/data/derived_data/sim_results.rds` (16{,}000 replicates across
363 32 scenarios). Driver scripts are at `analysis/scripts/`. ADEMP planning
364 documents are at `docs/plan-short.md` and `docs/plan-full.md`. The R package
365 source is at `R/`. The manuscript source, bibliography (`references.bib`), and
366 SIM CSL file are at `analysis/report/`.

367 11 References

368 11.1 Morris et al. (2019) ADEMP Compliance

369 We audited this simulation study against the reporting standards proposed by
370 Morris, White, and Crowther¹². The full audit is at `docs/morris-audit-2026-04-17.md`.

371 **Verdict:** Mostly compliant.

372 **Key gaps identified:**

- `n_sim = 500` not documented with an explicit Monte Carlo SE justification.
- `RNGkind('L\''Ecuyer-CMRG')` not pinned.
- Post-run `.Random.seed` not captured per replicate; coverage is not among the reported performance measures.

A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

Rendered on 2026-07-05 at 17:09 PDT.

Source: /Users/zenn/Library/CloudStorage/Dropbox/prj/res/17-mixed-r2/mixedr2/analysis/report/

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